

Exploring Design Directions for AI Teaching Assistants in Virtual Science Labs

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Abstract

Virtual science laboratories have gained traction in STEM education, yet replicating the real-time instructional support typically provided by human instructors remains a key challenge. To address this, we developed a virtual science lab integrating a GPT-4-powered AI teaching assistant (AI-TA) that delivers scripted instruction and responds to learner questions through a custom chat interface. This preliminary study explores learners' perceptions of the AI-TA, focusing on perceived benefits, limitations, and recommendations for future design. We conducted in-depth interviews and participatory design sessions with nine engineering students who had prior laboratory experience. Findings suggest that participants valued the AI-TA for its constant availability, reduced fear of judgment, and the increased engagement enabled by its 3D presence within the learning environment—benefits that go beyond traditional text-based chat systems. However, participants raised concerns about the accuracy of AI responses and the lack of seamless integration between chat interaction and instructional flow. Suggested improvements included smoother transitions between instruction and chat, context-aware support based on user activity, and more proactive engagement through structured AI prompts rather than relying solely on learner-initiated questions. These insights offer guidance for future development of AI-assisted virtual science labs.

CCS Concepts

• **Human-centered computing** → **Interaction Design**; *Empirical studies in interaction design*.

Keywords

Pedagogical Agent, AI Virtual Assistant, GPT, Virtual Science Lab

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1 Introduction

Virtual science laboratories have attracted growing interest as a promising approach to address the limitations inherent in traditional physical laboratories, namely high operational costs, physical space limitations, and safety concerns [2, 15]. Despite these advantages, one of the major challenges in virtual science labs lies in replicating the essential instructional support that human educators typically provide [15]. Gericke et al. [7] emphasized the crucial role of teacher support—such as responding promptly to student questions—in fostering students' conceptual understanding in laboratory-based education. Unlike physical labs, where instructors are available to provide immediate feedback and promote collaborative learning experiences, virtual environments often lack this real-time interaction and direct guidance. The rapid development and growing accessibility of large language models (LLMs), such as OpenAI's GPT, offer a new opportunity to bridge this gap [3, 19].

To explore this potential, we conducted a preliminary study investigating the integration of an AI teaching assistant into a virtual science laboratory. Our specific focus is on semiconductor fabrication and assurance—a field that has historically been difficult for many institutions to offer as a hands-on lab experience due to its dependence on highly specialized equipment, often costing millions of dollars, and the need for detailed operational knowledge typically delivered by expert-level teaching assistants [9, 16]. To this end, we developed a virtual lab that enables students to practice essential equipment operation procedures related to microelectronic fabrication, in collaboration with semiconductor engineers. Within this environment, the AI-powered assistant functions as a pedagogical agent, delivering step-by-step instructional narration to guide learners through the activities. To supplement this guidance, we integrated a custom GPT-powered chat interface—referred to as the “AI-TA”—which allows students to ask questions at any point during the tutorial. The system was specifically designed for desktop use to support greater scalability for larger student cohorts in future implementations.

This study focuses on two primary goals: (1) to examine engineering students' perceptions of using a GPT-powered AI teaching assistant in a virtual lab environment—including perceived benefits and limitations—and (2) to gather concrete design suggestions for improving our current prototype to better support student engagement and learning. To achieve these aims, we conducted in-depth interviews with engineering students who had prior experience with laboratory-based instruction. These interviews were followed by a participatory design session to elicit more detailed and actionable feedback [17]. This work seeks to contribute to the growing body of literature on intelligent tutoring systems in STEM education [1, 19, 20].

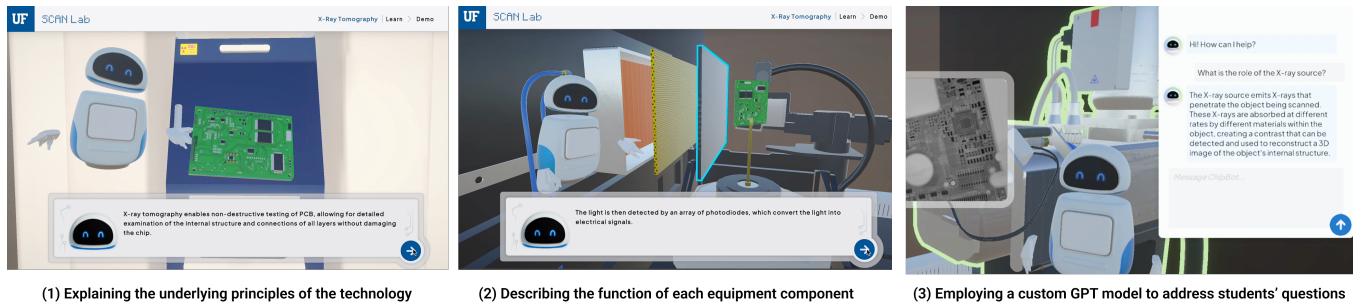


Figure 1: Screenshots of the virtual science lab: the TA guides the demonstration, while students can ask questions via the chat.

2 Related Works

Recent advances in large language models (LLMs) have sparked growing interest in their application as virtual tutors in educational contexts [6]. A systematic review by García-Méndez et al. [6] highlights that most existing implementations of LLMs in education focus on generating explanations, automating question creation, and providing assessment feedback through natural language dialogue. These applications have been especially prominent in mathematics and computer science, where AI tutors provide step-by-step explanations and adaptive feedback, such as validating formulas or debugging code. For instance, Yudelson and Brusilovsky’s NavEx system offers individualized code debugging along with illustrative examples [4], and Lane [10] developed PROPL, an AI tutor that assists students in constructing natural-language-style pseudocode solutions to programming problems. Despite these promising developments, most LLM-based educational tools remain limited to text-based environments, in which the AI tutor operates independently of the learner’s primary task [6, 8]. While this mode of interaction may be suitable for text-centric domains such as programming, it presents limitations in more interactive and visually rich domains, such as virtual science laboratories, where learning is often driven by spatial reasoning, visual cues, and procedural tasks [13, 14].

Only a limited number of studies have explored how LLMs can be integrated into immersive and interactive learning environments to support complex, task-based learning in the context of virtual science labs. Notably, Ward et al. developed My Science Tutor (MyST), an intelligent tutoring system embedded in a desktop-based environment aligned with the Full Option Science System curriculum [20]. More recently, Tracy and Spantidi [19] examined a GPT-powered conversational agent embedded within a virtual reality (VR) science classroom. Although these studies provide evidence for the potential of AI tutors in science education, these prior research has primarily focused on assessing the quantitative impact of AI tutors or virtual labs independently. Relatively little attention has been paid to qualitative investigations exploring how AI tutors can be effectively integrated into virtual science labs to function as instructional assistants. Embedding a virtual tutor within an interactive, task-driven environment presents unique design challenges—such as incorporating question-and-answer interactions without disrupting instructional flow, and ensuring that the AI agent’s support is contextually grounded in the learner’s ongoing activity. These challenges underscore the growing need for design-oriented research

that examines how conversational AI can be seamlessly embedded into interactive virtual learning environments.

3 Virtual Science Lab with AI-TA: Preliminary Design

3.1 Design Process

Designing a virtual lab for semiconductor fabrication and assurance requires interdisciplinary collaboration. To support this, we formed a team of semiconductor engineers, electrical engineering faculty, HCI researchers, UX designers, and software developers. Following Liu et al.’s [12] cross-disciplinary design framework for educational tools, the team began by defining learning objectives and standard lab procedures. To ensure authenticity, we reviewed recordings of real lab sessions to analyze instructional sequences. These insights guided the creation of user flow diagrams and interface mockups, aligning lab activities with the AI agent’s instructional prompts and chat-based support. Expert feedback was incorporated through iterative design cycles.

3.2 Learning Content and Interface

The virtual science lab is designed to teach the operation of micro-electronic fabrication equipment through structured, interactive tutorials. For instance, in the X-ray tomography module, learners are first introduced to the underlying principles of X-ray tomography technology. They are then guided through a step-by-step process involving tasks such as placing a chip on the testbed, aligning it, adjusting the voltage and current of the X-ray source, and interpreting the resulting images. The virtual lab interface is developed for desktop use, utilizing mouse and keyboard interactions. Users manipulate 3D objects—such as positioning the chip—through drag-and-drop actions and advance through the tutorial by clicking “Next” buttons. Instruction is delivered by a virtual teaching assistant via voice narration, accompanied by synchronized on-screen text. In addition to the scripted lecture, an AI-powered chatbox enables learners to ask the virtual assistant questions relevant to the instructional content. For example, while the assistant explains the relationship between voltage and current in the X-ray machine through pre-scripted dialogue, learners can independently ask questions such as, “What is voltage?” using the chat interface. An initial prototype is illustrated in Figure 1.

Table 1: Summary of Themes and Subthemes for AI-Powered TAs

Main Theme	Subthemes
Perceived Benefits of AI-Powered Virtual TAs	- Increased flexibility and availability compared to human TAs - Encouragement of questions by reducing fear of judgment - Improved efficiency in TA training for complex equipment operation - Viewed as knowledgeable and able to reduce human error when trained on domain-specific content
Added Value of the 3D AI Agent	- AI explanations paired with contextualized visuals support comprehension - Embedding the AI as a 3D avatar enhances its social presence
Potential Limitations of AI-Powered TAs	- Concerns about the accuracy and reliability of AI-generated responses - Prior negative experiences (e.g., AI customer service) contribute to skepticism
Design Recommendations for Future AI TAs	- Seamless integration of AI responses with instructional content - Enhanced context-aware interaction (e.g., object-specific feedback) - Use of inquiry-based prompts to foster deeper engagement

3.3 Apparatus

The virtual lab was developed using Unity Engine (version 2020.3.18). The AI teaching assistant was designed in the form of an “Eve-style” robotic avatar, a choice informed by previous research indicating that abstract, non-human designs can help reduce bias related to gender and appearance [11]. The assistant’s voice was generated via the ElevenLabs API, while conversational capabilities were powered by OpenAI’s GPT-4 API. A custom GPT agent was developed using OpenAI’s GPT Builder to ensure that the assistant could provide context-specific responses related to semiconductor fabrication procedures and equipment operations.

4 Exploring User Perspectives: Participatory Design

4.1 Overview and Procedure

As a preliminary evaluation, we adopted a two-phase approach to gather user feedback: in-depth interviews to capture initial impressions, followed by participatory design sessions to obtain more detailed input on interface and usability. In the first phase, participants engaged with the prototype to learn about X-ray tomography and were encouraged to ask questions using the AI chatbox. Interviews focused on three areas: perceived benefits of the AI chat, its limitations, and suggestions for improvement. Insights from this phase informed revisions to the prototype, which were then evaluated in a second session involving the same participants. During this session, participants offered more in-depth feedback and created sketches to illustrate their design ideas, as shown in Figure 2. All sessions were recorded, transcribed, and analyzed using thematic analysis [5]. Two researchers independently coded the transcripts using MAXQDA, then collaborated through axial coding to identify key patterns. Iterative discussions led to the synthesis of core themes. The summary of thematic analysis results from the interview findings is presented in Table 1.

4.2 Participants

Participants were recruited through convenience sampling and met one of the following criteria: (1) engineering students interested in

semiconductor topics but with no prior knowledge of the specific content presented in the prototype (representing the target learner group), or (2) engineering students with previous experience in lab-based sessions as teaching assistants. A total of nine participants took part in the study: seven Ph.D. students, one master’s student, and one undergraduate, ranging in age from 22 to 33 (Table 2). All participants had experience in laboratory-based learning environments, and three of them had previously served as teaching assistants in lab settings.

Table 2: Demographics of Participants

Participant ID	Education Background	Gender	Age
A1	Ph.D. (Chemical Engineering)	Male	27
A2	Ph.D. (Astronomy)	Male	33
A3	Ph.D. (Material Science Engineering)	Male	28
A4	Undergraduate (Electrical Engineering)	Male	22
A5	Ph.D. (Chemical Engineering)	Male	29
A6	Master (Electrical & Computer Engineering)	Male	28
A7	Ph.D. (Electrical & Computer Engineering)	Female	29
A8	Ph.D. (Chemical Engineering)	Male	29
A9	Ph.D. (Chemical Engineering)	Male	27

4.3 Initial Interview: Findings

4.3.1 Perceived Benefits of AI-Powered Virtual TAs. Participants repeatedly highlighted the AI TA’s flexibility and availability, noting that the chat interface provides support beyond traditional office hours. For example, one student remarked,

“With an AI agent, I can ask anything at any time—I’m not limited by office hours” (P6)

Learners also reported feeling more at ease asking questions, particularly those they might hesitate to raise in a live classroom:

“With an AI agent, I feel like I can ask anything without hesitation. If I ask a teacher ‘What is voltage?’ I might worry they’d think, ‘Why don’t you know this?’ But with AI, I don’t have to worry.” (P1)

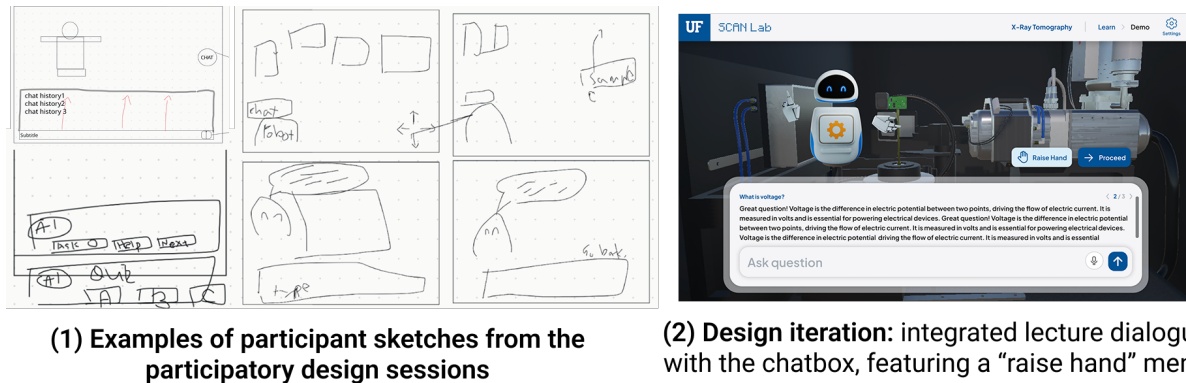


Figure 2: Examples from participatory design sessions(left), and the iterated design developed based on feedback. (right)

Several participants observed that the virtual lab could streamline TA training by reducing the time and effort required to onboard new human assistants:

"Training TAs on how to use equipment takes a lot of time. This system could help save that." (P8)

Finally, some learners perceived the AI as more knowledgeable than human TAs when it is trained on domain-specific content:

"While TAs try to explain, sometimes they don't know the answer. But this app is really great in that sense." (P4)

4.3.2 Added Value of the 3D AI Agent in the Virtual Lab. Participants reported that embedding the AI within a 3D virtual environment improved their understanding of complex equipment operations by combining verbal explanations with contextualized visuals:

"GPT explains things well in text, but it's sometimes hard to fully understand without visuals. What I liked about this was how the visuals made complex ideas easier to grasp. For basic questions like 'What does PCB stand for?' text is fine, but for something like 'How is X-ray tomography used and why does it matter?', visuals really help clarify the process and potential issues." (P3)

Three participants further noted that the presence of a 3D character increases engagement:

"Instead of just hearing a voice, a cute robot appeared and spoke, making it feel more like it was talking to me. That might be why it felt more engaging." (P3)

4.3.3 Perceived Limitations of AI-Powered TAs. Despite these advantages, several learners expressed concern about the AI's accuracy, particularly when addressing emerging or advanced topics:

"ChatGPT's accuracy often drops with newer technologies due to data limitations. It would be more reliable if tailored to course materials provided by professors in university classes." (P4)

Moreover, prior negative experiences with rudimentary chatbots in many customer service websites led some to approach the AI-TA with skepticism:

"Those chatbots on school websites usually can't answer anything. This one felt smarter, probably because it's based on ChatGPT. It would help if that (GPT) were made clear to set it apart." (P2)

4.3.4 Design Recommendations for Future AI-Powered TAs. Participants advocated for tighter and seamless integration between AI responses and the planned instructional sequence:

"I liked the chat feature—it helped when I forgot the voltage-current relationship. But I assumed it would follow the lesson content, and it didn't. That's when I realized I needed to ask more specific questions." (P1)

They also recommended the development of context-aware interactions that respond to the learner's current view within the simulation:

"The app explains things verbally, right? But I'd like to ask about specific parts while viewing the images. If I could select a region and the AI explained what it is and does, that would be really helpful." (P3)

Finally, several participants suggested that the AI could proactively support inquiry-based learning by posing reflective questions or interactive challenges:

"I think GPT could make learning more engaging by asking thought-provoking questions. For example, during training, it could show an X-ray and ask, 'The contrast is poor what would you adjust?' I've seen similar techniques on lab tours, where guides ask questions to keep students focused. If the chatbot included short quizzes or follow-ups like, 'That's close, but actually...' it will help with retention." (P3)

4.4 Follow-up Participatory Design Session: Findings

Building on insights gathered from the in-depth interviews, a follow-up participatory design session was conducted to collect more specific, actionable recommendations for improving the AI teaching assistant interface. During this session, participants offered detailed design suggestions and created sketches to visualize their ideas, as illustrated in Figure 2.

Importance of Chat History for Learning Participants emphasized the value of being able to view past interactions with the AI assistant, particularly when dealing with complex scientific terminology. As one participant explained:

"With complex scientific terminology, sometimes it's hard to remember exactly what to ask, especially if the terms are unfamiliar or hard to recall. So having the conversation history visible would help." (P5).

To address this need, P5 proposed adding a menu that allows users to expand and review the chat history within the interface.

Mid Check-Ins and Structured Prompts Another participant highlighted the potential of using structured prompts and periodic check-ins to guide learners through the module and encourage meaningful interaction with the AI assistant:

"Because if people just read on their own without interacting or failing at something, they might not even know what to ask. But if there's some structure—like, the robot starts with Equipment 1, and then goes through a series of steps—it can be kind of like a pre-scripted game." (P8)

The same participant also suggested incorporating short quizzes at key points in the module to assess comprehension and keep learners engaged throughout the session.

Transitions Between Lecture and AI Conversation Smooth transitions between the lecture and Q&A modes were identified as another area for improvement. One participant recommended implementing features that simulate classroom dynamics, such as a "raise hand" option to initiate a question. Additionally, P3 proposed using wrap-up phrases to help re-establish the instructional flow after a question is answered: "That was a great question. Let's continue." This type of phrasing, they suggested, would support a more natural and cohesive learning experience, allowing students to transition more fluidly between asking questions and following instructional content.

5 Conclusion

This preliminary study seeks to explore design strategies for integrating a GPT-powered AI teaching assistant into a virtual microelectronic fabrication lab to enhance instructional support. We designed our initial prototype and conducted interviews and participatory design sessions with engineering students who had prior lab experience, either as students or teaching assistants. Participants valued the AI-TA for its constant availability, reduced social pressure compared to human TAs, and the enhanced sense of teacher presence enabled by its 3D embodiment. The addition of a natural language chat interface allowed learners to ask questions, while the scripted voice guidance helped structure the overall learning experience. However, participants expressed concerns about the scientific accuracy of AI-generated responses and the lack of seamless integration between instructional flow and conversational interactions in the current prototype. Through participatory design activities, students proposed a range of improvements, including embedded quizzes, access to chat history, context-aware tutor support based on user actions, and smoother transitions between instruction and question-answering modes. In alignment with prior work, these

findings underscore the need for AI agents that are both contextually aware and pedagogically aligned with the learning environment [18, 21]. As large language models continue to evolve, future research could explore how these systems can more effectively respond to learners' real-time actions and become more deeply integrated into the procedural workflows of virtual labs.

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References

- [1] Rania Abdelghani, Yen-Hsiang Wang, Xingdi Yuan, Tong Wang, Pauline Lucas, Hélène Sauzéon, and Pierre-Yves Oudeyer. 2024. GPT-3-driven pedagogical agents to train children's curious question-asking skills. *International Journal of Artificial Intelligence in Education* 34, 2 (2024), 483–518. doi:10.1007/s40593-023-00340-7
- [2] Murat Akçayır, Gökçe Akçayır, Hüseyin Miraç Pektaş, and Mehmet Akif Ocak. 2016. Augmented reality in science laboratories: The effects of augmented reality on university students' laboratory skills and attitudes toward science laboratories. *Computers in Human Behavior* 57 (2016), 334–342. doi:10.1016/j.chb.2015.12.054
- [3] Anissa M Bettayeb, Manar Abu Talib, Al Zahraa Sobhe Altayasinah, and Fatima Dakalbab. 2024. Exploring the impact of ChatGPT: conversational AI in education. In *Frontiers in Education*, Vol. 9. Frontiers Media SA, 1379796. doi:10.3389/educ.2024.1379796
- [4] Peter Brusilovsky and Michael V Yudelson. 2008. From webex to navex: Interactive access to annotated program examples. *Proc. IEEE* 96, 6 (2008), 990–999.
- [5] Victoria Clarke and Virginia Braun. 2017. Thematic analysis. *The journal of positive psychology* 12, 3 (2017), 297–298.
- [6] Silvia García-Méndez, Francisco de Arriba-Pérez, and María del Carmen Somoza-López. 2024. A review on the use of large language models as virtual tutors. *Science & Education* (2024), 1–16. doi:10.1007/s11191-024-00530-2
- [7] Niklas Gericke, Per Höglström, and Johan Wallin. 2023. A systematic review of research on laboratory work in secondary school. *Studies in science education* 59, 2 (2023), 245–285. doi:10.1080/03057267.2022.2090125
- [8] Wayne Holmes and Ilkka Tuomi. 2022. State of the art and practice in AI in education. *European journal of education* 57, 4 (2022), 542–570.
- [9] Hyo Kang, Serene Cheon, Alice Abia-Okon, Aida Damanpak, Wanli Xing, and Navid Asadizanjani. 2024. Addressing the Talent Gap in Semiconductors: Motivators and Barriers to Career Choices. In *2024 IEEE Frontiers in Education Conference (FIE)*. IEEE, 1–5. doi:10.1109/FIE61694.2024.10893357
- [10] H Chad Lane and Kurt VanLehn. 2005. Teaching the tacit knowledge of programming to novices with natural language tutoring. *Computer Science Education* 15, 3 (2005), 183–201. doi:10.1080/08993400500224286
- [11] Jamy Li, René Kizilcec, Jeremy Bailenson, and Wendy Ju. 2016. Social robots and virtual agents as lecturers for video instruction. *Computers in Human Behavior* 55 (2016), 1222–1230. doi:10.1016/j.chb.2015.04.005
- [12] Chang Liu, Felicia Fang-Yi Tan, Shengdong Zhao, Abhiram Kanneganti, Gosavi Arundhati Tushar, and Eng Tat Khoo. 2024. Facilitating Virtual Reality Integration in Medical Education: A Case Study of Acceptability and Learning Impact in Childbirth Delivery Training. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*. 1–14. doi:10.1145/3613904.3642100
- [13] Roxana Moreno and Richard E Mayer. 2002. Learning science in virtual reality multimedia environments: Role of methods and media. *Journal of educational psychology* 94, 3 (2002), 598. doi:10.1037/0022-0663.94.3.598
- [14] Jocelyn Parong and Richard E Mayer. 2018. Learning science in immersive virtual reality. *Journal of educational psychology* 110, 6 (2018), 785. doi:10.1037/edu0000241
- [15] Veljko Potkonjak, Michael Gardner, Victor Callaghan, Pasi Mattila, Christian Guetl, Vladimir M Petrović, and Kosta Jovanović. 2016. Virtual laboratories for education in science, technology, and engineering: A review. *Computers & Education* 95 (2016), 309–327. doi:10.1016/j.compedu.2016.02.002
- [16] Aida Damanpak Rizi, Antika Roy, Rouhan Noor, Hyo Kang, Nitin Varshney, Katja Jacob, Sindia Rivera-Jimenez, Nathan Edwards, Volker J Sorger, Hamed Dalir, et al. 2023. From talent shortage to workforce excellence in the CHIPS Act era: Harnessing industry 4.0 paradigms for a sustainable future in domestic chip production. *arXiv preprint arXiv:2308.00215* (2023).
- [17] Douglas Schuler and Aki Namioka. 1993. *Participatory design: Principles and practices*. CRC press.
- [18] Philipp Spitzer, Joshua Holstein, Patrick Hemmer, Michael Vössing, Niklas Kühl, Dominik Martin, and Gerhard Satzger. 2025. Human Delegation Behavior in Human-AI Collaboration: The Effect of Contextual Information. *Proc. ACM Hum.-Comput. Interact.* 9, 2, Article CSCW101 (May 2025), 28 pages. doi:10.1145/3710999

- [19] Kaitlyn Tracy and Ourania Spantidi. 2025. Impact of GPT-Driven Teaching Assistants in VR Learning Environments. *IEEE Transactions on Learning Technologies* 18 (2025), 192–205. doi:10.1109/TLT.2025.3539179
- [20] Wayne Ward, Ron Cole, Daniel Bolaños, Cindy Buchenroth-Martin, Edward Svirsky, and Tim Weston. 2013. My science tutor: A conversational multimedia virtual tutor. *Journal of Educational Psychology* 105, 4 (2013), 1115. doi:10.1037/a0031589
- [21] Songlin Xu, Hao-Ning Wen, Hongyi Pan, Dallas Dominguez, Dongyin Hu, and Xinyu Zhang. 2025. Classroom Simulacra: Building Contextual Student Generative Agents in Online Education for Learning Behavioral Simulation. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. Association for Computing Machinery, New York, NY, USA, Article 789, 26 pages. doi:10.1145/3706598.3713773